NEUROLOCK: REAL-TIME INTRUSION DETECTION SYSTEM (IDS) IN C++

Abstract

NeuroLock is a high-performance, real-time Intrusion Detection System (IDS) developed in C++ using libpcap. It continuously monitors network traffic, identifies port scans, ICMP floods, and suspicious application-layer payloads, and raises alerts in real time. This document details the design, implementation, and testing of NeuroLock in a controlled, safe environment on localhost, demonstrating its ability to detect both network-layer and application-layer attacks while providing actionable insights for cybersecurity monitoring

KARABO MANUEL KAMOGELO MOLAPO

Molapomanuel709@gmail.com

Project overview

NeuroLock is a high-performance, real-time Intrusion Detection System (IDS) built in C++ using libpcap. It captures live network traffic, analyzes packet content, detects common attacks.

Key Features:

- Real-Time Packet Capture – Continuously captures live network traffic on the host machine.
- Port Scan Detection – Detects aggressive scanning activity by tracking connection attempts across multiple ports.
- ICMP Flood Detection – Identifies potential DoS attacks using ICMP traffic analysis.
- Payload Inspection – Monitors TCP payloads for suspicious keywords indicative of attacks (e.g., admin, password, SQL injection).
- Lightweight and Fast – Optimized for low-latency packet processing using C++ and efficient data structures (map, set)

How It Works

NeuroLock operates by capturing packets from the network interface using libpcap. Only relevant traffic, specifically TCP and ICMP packets, is processed to maintain performance. Each packet is parsed layer by layer: the Ethernet header identifies IP traffic, the IP header extracts source and destination addresses, and the TCP or ICMP header determines the type of traffic. The IDS implements detection logic for port scans, ICMP floods, and suspicious payloads. Port scans are detected by tracking multiple port access attempts from a single IP within a short time. ICMP floods are flagged when packet counts exceed a threshold, indicating potential DoS attacks. Payload inspection identifies suspicious strings in TCP payloads to catch application-layer attacks

Port Scan Detection

Test Environment:

- Localhost (127.0.0.1) on my machine
- Python HTTP server running on port 8080 to simulate a live service

Purpose:

- To test if NeuroLock IDS can detect multiple ports being scanned in succession, which simulates a common reconnaissance attack.

Test Procedure:

Start the HTTP server on port 8080:

```
~ via 🐍 v3.13.12
❯ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

Start NeuroLock IDS (Safe Mode – monitors only localhost):

- Run a full TCP port scan from 1 to 9000 on localhost:

```
~ via 🐍 v3.13.12
❯ nmap -sT -p- 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 02:27 +0200
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000055s latency).
Not shown: 65530 closed tcp ports (conn-refused)
PORT      STATE SERVICE
8000/tcp open  http-alt
8065/tcp open  unknown
8080/tcp open  http-proxy
8089/tcp open  unknown
8191/tcp open  limnerpressure

Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds
```

Observed Behavior:

- NeuroLock detected the scan immediately when 3 or more ports were hit:

```
~ via 🐍 v3.13.12
❯ sudo ./neurolock
[sudo] password for manuel:
[*] Using device: lo
[*] NeuroLock IDS Running (Safe Mode: ON - localhost only)...
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] ICMP Flood detected from 127.0.0.1 targeting 127.0.0.1
```

Interpretation:

- The IDS correctly identified suspicious scanning activity even though it was performed on the same machine.
- This demonstrates the IDS's ability to monitor network activity in real-time and raise alerts before a potential compromise.

CMP Flood Detection

Purpose:

- To test NeuroLock's ability to detect flooding attacks, which are often used in DoS (Denial-of-Service) scenarios.

Test Procedure:

Use a high-frequency ping to localhost:

```
~ via 🐍 v3.13.12 took 2s
❯ sudo ping -i 0.002 -f 127.0.0.1
[sudo] password for manuel:
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
```

2. Monitor NeuroLock IDS for alerts.

Observed Behavior:

- NeuroLock raised an alert once the ICMP packet threshold was exceeded:

```
Session  Actions  Edit  View  Help
❯ sudo ./neurolock
[sudo] password for manuel:
[*] Using device: lo
[*] NeuroLock IDS Running (Safe Mode: ON - localhost only)...
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] ICMP Flood
```

IDS: NeuroLock monitoring 127.0.0.1 only (Safe Mode)

Purpose

- To test NeuroLock's ability to detect suspicious HTTP POST requests, simulating a login or brute-force attempt.
- Ensures detection of application-layer attacks, not just network traffic.

Test Procedure

- Send a POST request with login-like data to the Flask server:

```
manuel@kali: ~
Session  Actions  Edit  View  Help

~ via 🐍 v3.13.12
❯ curl -X POST http://127.0.0.1:8080 -d "username=admin&password=123"
OK

~ via 🐍 v3.13.12
```

Observed behavior:

```
~ via 🐍 v3.13.12
❯ sudo ./neurolock
[sudo] password for manuel:
[*] Using device: lo
[*] NeuroLock IDS Running (Safe Mode: ON - monitoring 127.0.0.1)...
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Port scan detected from 127.0.0.1 targeting 127.0.0.1
[ALERT] Suspicious POST payload detected from 127.0.0.1 to 127.0.0.1
```

Interpretation

- It detected the POST payload because it contained keywords like username= and password=.
- The IDS flagged it as suspicious to demonstrate that it can analyze application-layer traffic, not just TCP/ICMP headers.
- The alert includes source IP, destination IP,type of payload.

NeuroLock IDS Summary

NeuroLock IDS was tested in a controlled environment on localhost. It successfully detected port scans, ICMP floods, and suspicious POST payloads, logging detailed alerts in real time. All tests were safe, legal, and confined to the local machine, making it an ideal demonstration for portfolios or LinkedIn. The project showcases NeuroLock's ability to provide real-time, full-stack network security monitoring, combining both network-layer and application-layer intrusion detection.